

Dexterous Piano Track Milestone

Baseline Reproduction and Next Steps

Sining Zhoubian Kun Feng Xingyu Shen Wenkai Li Jian Gao

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- Piano playing requires long-horizon planning, precise contact timing, and coordinated bimanual dexterity.
- We build on PianoMime, which combines internet demonstration imitation with song-specific residual reinforcement learning.
- Milestone goal: reproduce the original baseline faithfully and prepare a reliable pipeline for improvement experiments.

Problem Formulation

- Environment: RoboPianist with two Shadow Hands and an 88-key digital piano.
- Goal: play a MIDI-specified musical piece with correct key presses and timing.
- Metrics: note-level precision, recall, and F1 from the MIDI evaluation wrapper.
- Two policy families:
 - single-song residual PPO with demonstration trajectory + IK/QP prior;
 - generalist high-level + low-level diffusion checkpoint evaluation.

Single-song policy

- Uses stored song demonstration trajectories.
- Computes an IK/QP prior action.
- PPO learns a residual correction.
- Does not load diffusion checkpoints.

Generalist policy

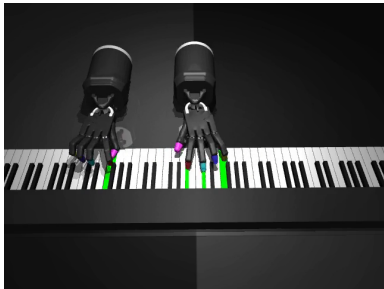
- Uses released high-level diffusion checkpoint.
- Predicts fingertip trajectories.
- Uses low-level diffusion to generate robot actions.
- Evaluated on unseen test songs.

Single-Song Replay Baseline

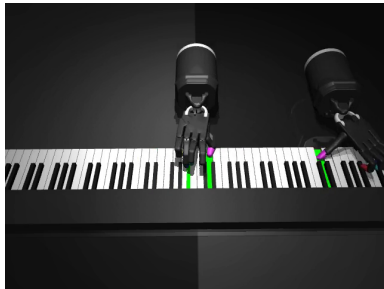
Song	Precision	Recall	F1
Stan_1	0.9991	0.9719	0.9795
Petrunko_3	0.9869	0.8460	0.8900
NeverGonnaGiveYouUp_1	0.9960	0.9260	0.9514

Videos: [baseline_results/single_song/videos/](#)

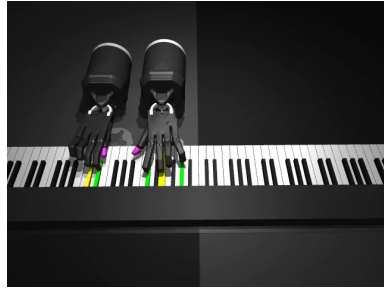
Single-Song Visual Examples



Stan_1

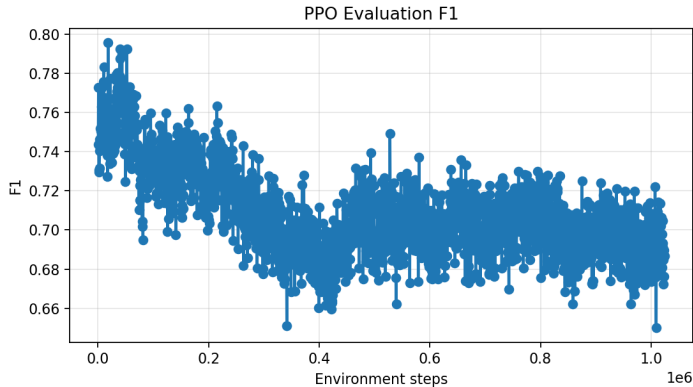


Petrunko_3



NeverGonnaGiveYouUp_1

Residual PPO Training Curve

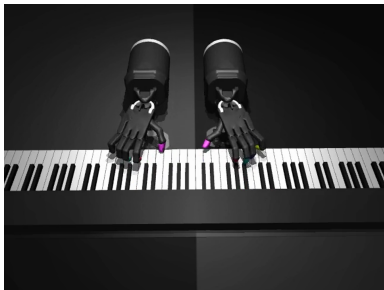


- Song: Petrunko_3
- Iterations: 2000
- Environment steps: 1,024,000
- Best-checkpoint rollout F1: 0.7957
- Last evaluation F1: 0.6867

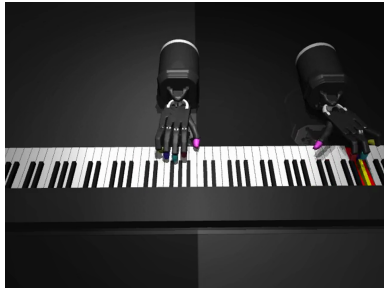
Generalist Diffusion Baseline

Song	Precision	Recall	F1
Alone_1	0.8283	0.6443	0.7902
Numb_1	0.5286	0.4521	0.7504
NoTimeToDie_1	0.8192	0.8076	0.8553
Forester_1	0.8116	0.7300	0.7944
EyesClosed_1	0.6127	0.5151	0.8569
Paradise_1	0.8392	0.7535	0.8104
SomewhereOnlyWeKnow_1	0.6516	0.5789	0.7920

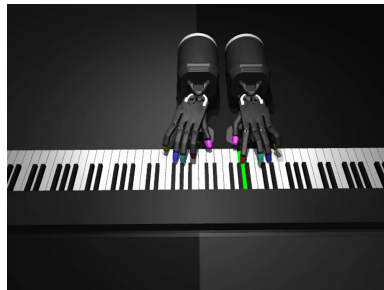
Generalist Visual Examples



NoTimeToDie_1



EyesClosed_1



Paradise_1

Single-song videos

- `/home/gaoj/share4/_piano/baseline_results/single_song/videos/`

Generalist videos

- `/home/gaoj/share4/_piano/baseline_results/multisong/videos/`

PPO rollout video

- `.../Petrunko_3_ppo_curve_20260513_135059/eval/02001.mp4`

Note: current videos are silent due to missing system FluidSynth/PortAudio libraries; visual simulation and note-level metrics are valid.

Next Steps

- Single-song improvement: select a difficult/long training piece and compare baseline vs improved F1 curve and play-through video.
- Generalist improvement: compare baseline and improved F1 on at least five unseen songs.
- Candidate directions:
 - reward shaping for timing, contact, and smoothness;
 - residual-aware imitation;
 - contact-centric goal representation;
 - RL tricks such as RSI, tempo curriculum, and residual regularization.